

Vespa is an open source, scalable Big Data serving platform that supports efficient storage and retrieval of large-scale vector embeddings. With built-in support for vector indexing and similarity search functionalities, Vespa enables LLMs and chatbots to efficiently query and retrieve relevant information from vast collections of textual data. Features such as real-time updates, ranking models, and distributed processing capabilities make it well-suited for building high-performance AI applications with stringent latency and throughput requirements.

Milvus is yet another vector database that has gained traction in the LLM and AI chatbot community. An open source vector similarity search engine developed by Zilliz, it is designed to handle massive-scale vector indexing and retrieval tasks efficiently. Leveraging cutting-edge indexing algorithms and data structures optimised for vector data, Milvus offers fast and accurate similarity search capabilities, enabling LLMs and chatbots to process large volumes of textual data with minimal latency. It supports flexible deployment options, including standalone, distributed, and cloud-based deployments.

Other open source libraries and frameworks such as FAISS (Facebook AI Similarity Search) and Annoy (Approximate Nearest Neighbors Oh Yeah) provide essential tools and algorithms for vector indexing and similarity search tasks. FAISS, developed by Facebook AI Research, offers highly optimised implementations of state-of-the-art algorithms for efficient similarity search in large-scale vector spaces. Similarly, Annoy, developed by Spotify, provides a lightweight and efficient library for approximate nearest neighbour search, making it suitable for resource-constrained environments and rapid prototyping of AI applications.

Free datasets for LLMs and AI chatbots

The realm of artificial intelligence, particularly in the development and deployment of large language models (LLMs) and AI chatbots, has witnessed an unprecedented surge in interest and innovation. Central to this burgeoning field is the availability of expansive, high-quality datasets which underpin the training and refinement of these sophisticated models. In the open source community, several free databases have emerged as pivotal resources, each offering unique attributes that cater to the diverse requirements of AI researchers and developers.

One of the most prominent free databases is the Common Crawl corpus, a repository that offers petabytes of web-crawled data amassed from billions of web pages. This dataset is invaluable for LLMs due to its vastness and diversity,

encompassing a myriad of languages and topics. The Common Crawl's extensive repository enables models to learn nuanced linguistic patterns and broad contextual understanding, essential for generating human-like text and engaging in coherent conversations. The corpus is updated regularly, ensuring that the data reflects the evolving nature of internet content, thereby enhancing the temporal relevance and accuracy of the models trained on it.

Another critical resource is the Wikipedia dataset, which, being more curated and structured than Common Crawl, provides a rich source of factual and encyclopaedic knowledge. The text from Wikipedia is meticulously written and edited, making it an excellent training ground for AI models that require a high degree of reliability and accuracy in their responses. The structured nature of Wikipedia data, with its comprehensive categorisation

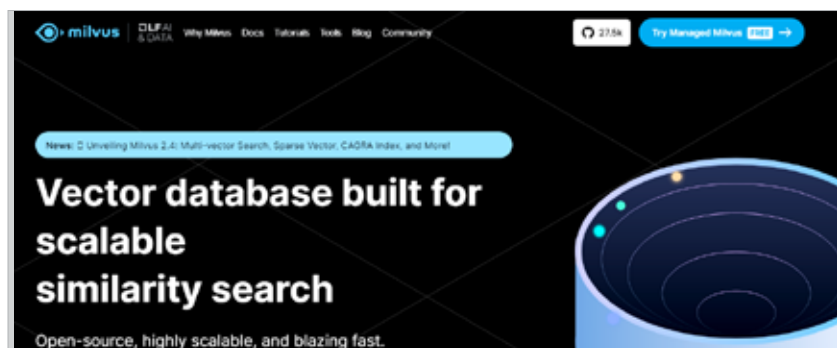


Figure 2: Milvus vector database

Prominent vector databases for AI chatbots

Database	URL
Milvus	https://milvus.io/
Pinecone	https://www.pinecone.io
Vespa	https://vespa.ai
Qdrant	https://qdrant.tech
Weaviate	https://weaviate.io
FAISS (Facebook AI Similarity Search)	https://github.com/facebookresearch/faiss
Annoy (Approximate Nearest Neighbors Oh Yeah)	https://github.com/spotify/annoy
Vald	https://vald.vdaas.org